VISHAL CHAUDHARY

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SUMMARY

B.Tech Information Technology student at NSUT with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Generative AI. Built end-to-end AI applications including RAG systems, AI agents, GraphRAG with Neo4j, Chat-with-SQL, and Computer Vision projects. Proficient in Python, LangChain, MCP, and modern LLM frameworks

SKILLS

Languages: Python, C++, SQL

Machine Learning: Classification, Regression, CNNs, Transfer Learning, Feature Engineering

Deep Learning: TensorFlow, Keras, Neural Networks

Generative AI: LangChain, RAG, GraphRAG, MCP, Prompt Engineering, AI Agents, HuggingFace

Databases: SQLite, Neo4j

Tools: Git, GitHub, Jupyter Notebook, Google Colab

EDUCATION

Netaji Subhas University of Technology (NSUT), New Delhi, India,
Expected Graduation: 2028
B.Tech in Information Technology
CGPA: 7.88/10

PROJECTS

- **GraphRAG with Neo4j -** Developed a GraphRAG system using Neo4j and LangChain for graph-based knowledge retrieval. Created knowledge graph relationships to improve contextual reasoning. Integrated graph retrieval with LLM-powered question answering.
  Tech: Python, Neo4j, LangChain, GraphRAG
- **AI Agent with Tool Calling -** Built an autonomous AI agent capable of reasoning and invoking external tools. Implemented multi-step workflows for complex task execution.
  Integrated APIs and tool-calling mechanisms.
  Tech: Python, LangChain, MCP, Ollama, Groq
- **PDF RAG Application -** Developed a Retrieval-Augmented Generation system for querying PDF documents. Implemented document chunking, embeddings, vector retrieval, and answer generation. Improved response accuracy using context-aware retrieval.
  Tech: Python, LangChain, RAG, HuggingFace
- **Chat with SQL Database -** Built a natural-language-to-SQL application allowing users to query databases conversationally. Generated SQL queries from user prompts and returned structured results.
  Tech: Python, SQLite, LangChain, Streamlit

- **YOLO-Based Vehicle Counting System -** Developed a real-time vehicle detection and counting system using YOLO and OpenCV. Tracked vehicles crossing a predefined counting line in video streams. Processed traffic footage to monitor vehicle flow and generate count statistics. Improved counting accuracy through object tracking and duplicate detection handling.
  Tech: Python, YOLO, OpenCV, Computer Vision

ACHIEVEMENTS

- Solved **200+ Data Structures and Algorithms problems** on **LeetCode.**
- Built **10+ AI/ML, Computer Vision, and Generative AI projects** including RAG, GraphRAG, AI Agents, MCP, and YOLO-based applications.
- Developed end-to-end applications using **LangChain, Neo4j, OpenCV, MediaPipe**
- Hands-on experience with both **Computer Vision** and **Large Language Model (LLM)** systems.